

CROP DISEASES

Presented by

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Introduction

- Crop diseases are those conditions when pathogens, lack or inadequacy of nutrient supply and various other factors of the environment cause disruptions or hinderances to the normal and natural processes of plant growth, development and reproduction (Udo and Ndon, 2016).

- The major factors responsible for crop diseases are fungi, bacteria, viruses and nematodes.
- Others are nutrient deficiencies and air pollutants.
- The severity of crop diseases can range from mild leaf or fruit damage to death.

Disease Identification

- Modifications brought about by the disease are called symptoms.
- Structural modifications provide guide to the identification of the disease type of the causative organism.

Hypertrophy

- o This refers to the enlargement of organs as a result of accelerated multiplication of cells that are under attack.
- o This causes tumours as is the case in root knots of tomato caused by *Scutellima brady* or *Meloidogyne incognita* (root knot nematode).

Necrosis

- o This symptom appears as dead patches or spots all over the leaves of affected plants. It leads to distortion and death of organs, especially the leaves.
- o It is caused by the attacks of fungi, bacteria and viruses.

Chlorosis

- This is a discolouration of tissues arising from nutrient deficiency or pathogenic attacks which destroys the chloroplasts and reduce chlorophyll quality.

Deformation of organs

- A lot of pathogens often cause permanent damages to the organs that they attack.
- Examples are the Cassava mosaic Virus (CMV) on cassava leaves or zinc deficiency on maize leaves and shoots.

Disease Control

◦ Physical methods

- ***Exclusion***: This involves preventing the disease from getting into a new farm or crop area. This includes plant quarantine and the use of disease-free planting materials.

- ***Avoidance***: In this case crops are grown in a manner that avoids exposing them to the disease.

For instance, early sowing, selection of farm site free from certain diseases, use of resistant varieties, suitable crop rotation etc.

Other physical methods

- An example is removing and burning or burying disease contaminated plant residues.

Biological methods

- o This involves the use of living organisms (beneficial organisms) to attack the disease causing organism

Chemical methods

- This involves the use of chemicals of different formulation to kill off the causative organism

Maize

- Disease: Leaf blight
- Causative organism: Fungi
(*Helminthosporium spp*)

Symptoms

Symptoms appear first as numerous white specks on the **leaves**, and also as spots that expand causing browning of the **leaves** that later die from the tips.



Management

- o : Crop rotation with a non host crop like legumes to reduce the disease severity.
- o Destruction of crop debris.
- o Ploughing under of straw.
- o **Neem seed extract.**

Maize

- Disease: Downy mildew
- Causative organism: Fungi
(*Peronosclerospora spp*)

Symptoms

mottling, chlorotic streaking and lesions, and white striped leaves that eventually shred



Management

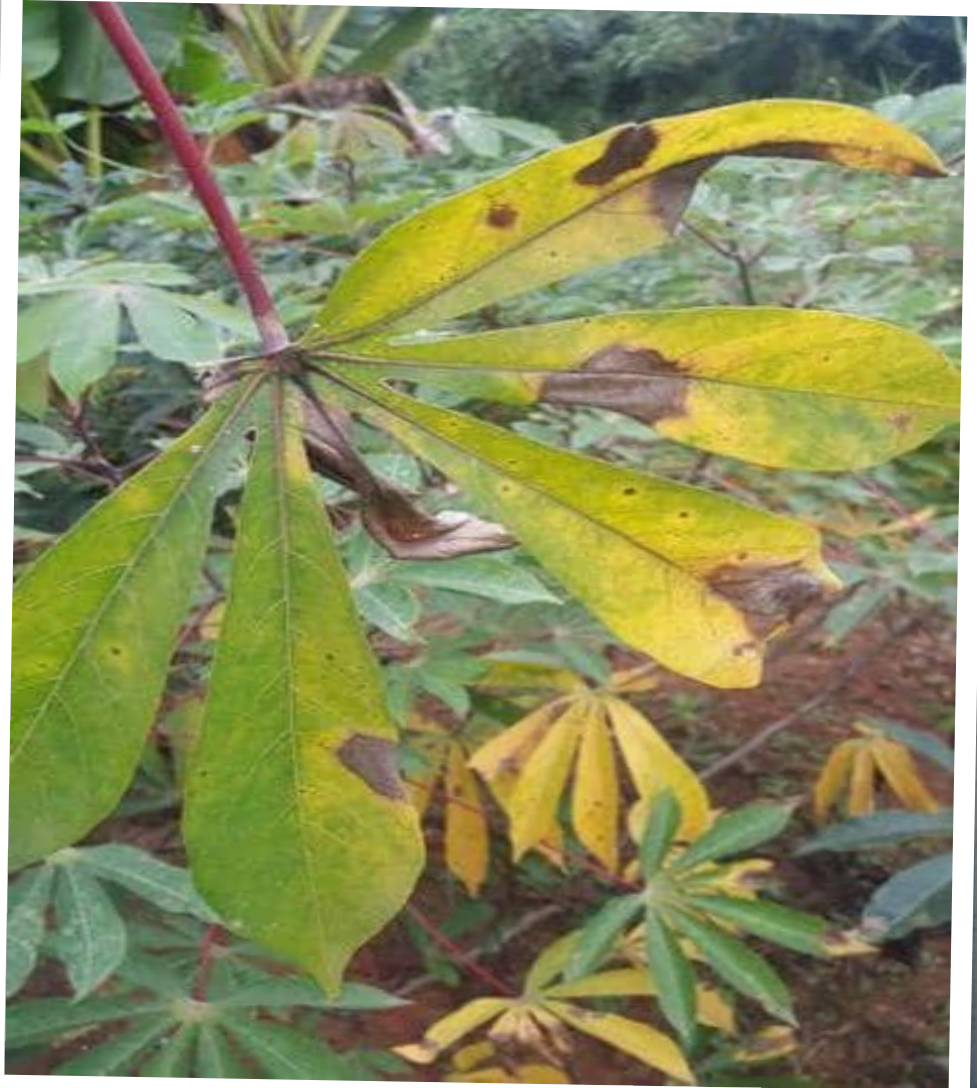
- o use of resistant cultivars.
- o Use disease free healthy seeds.
- o Use of Systemic Fungicides
Metalaxyl and Fosetyl-Al

Cassava

- Disease: Cassava bacterial blight
- Causative organism: *Xanthomonas manihotis*

Symptoms

Early **symptoms** appear as brown to dark-brown water-soaked angular spots on the **leaf** tissue browning at later stages



Management:

- Use of resistant varieties.
- Practice crop rotation. Cut out infected parts if only few.

Cassava

- Disease: Cassava brown leaf spot
- Causative organism: *Cercospora henningsii*

Symptoms

- Spots are produced on both the sides of leaves.
- On upper surface brown spots with dark border appears.
- On lower side of the leaf spots with grey centre and less distinct margin appear. The infected leaves turn yellow and dry up.



Management:

- Use of resistant varieties.
- During the dry season **remove heavily diseased leaves** to prevent the fungus spreading quickly when the rains start.
- After harvest: Collect and burn debris. Use fungicides where necessary

Cassava

- ◊ Disease: Cassava Mosaic Virus (CMV) disease
- ◊ Causative organism: Transmitted by white fly (*Bemisia tabaci*)

Symptoms

mosaic, mottling,
misshapen and
twisted leaflets,
and an overall
reduction in size of
leaves and plants



Management

- o Use of disease free planting materials.
- o The uprooted plants should be carried away from the field and exposed to the sunlight for drying and then burned to kill the **viruses**.
- o Do not plant cassava close to infected fields as the insect vectors will transport the disease to healthy plants.



o Thank You for
Listening...